LGLMF: **L**ocal **G**eographical based **L**ogistic **M**atrix **F**actorization Model for POI Recommendation

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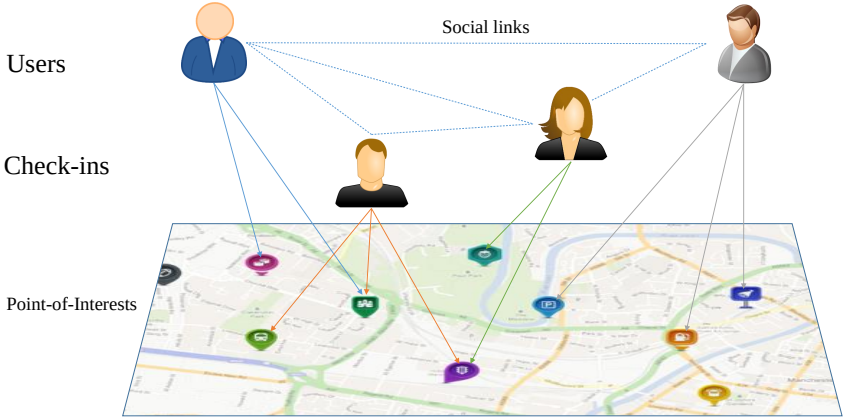
POI Recommendation



Sparsity and Context Information

- Data Sparsity
  - In most cases the density of check-in data is less than 0.05%
- Implicit Feedback
  - Users do not explicitly denote "like" or "dislike" by giving different scores! Check-in is just the frequencies of visits
- Context Information
  - Contextual information can alleviate data sparsity but how to incorporate different types of context information is another challenge!
  - {Geographical, Temporal, Social, Category, Comments, etc}

Geographical Influence

- One of the most important contextual factor
- Different methods just consider user's prospective
- Simple idea: People tend to visit POIs close to their location

Geographical models in POI recommendation:

1. Power-law Distribution
2. Multi-Center Gaussian
3. iGLSR
4. L-WMF

Motivation

- Many location-based social networks (LBSNs) have emerged in recent years
- To model a user based on her Geographical influence
- To model the user based on her history of preferences
- Jointly model of both users' and locations' point of view
- To propose a flexible geographical model

Proposed approach

Local Geographical Model

Context information, POI Locality, Users' and POIs' prospective

Logistic Matrix Factorization

Users' check-in history and feedback, Implicit Feedback property

A Unified Geographical Matrix Factorization Model

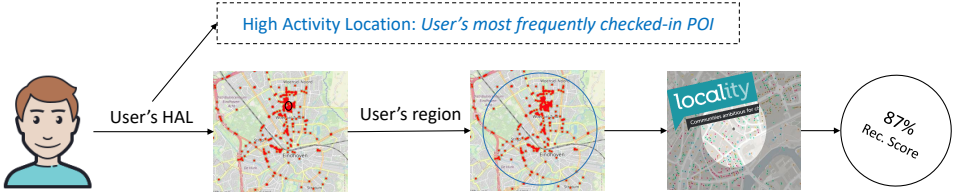
Data Sparsity

Local Geographical Model (LGM)

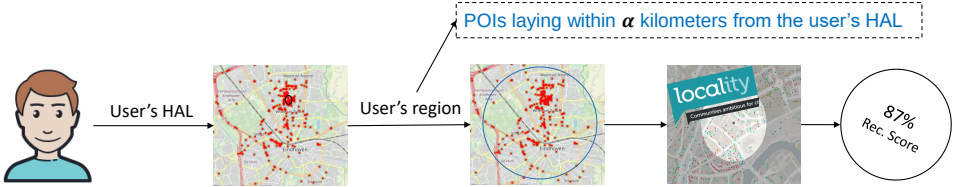
- Geographical model to capture the geographical influence of both users' and locations' points of view
- From the user's point of view, the geographical information can be modeled by considering the user's activity region
- From the location's point of view, the geographical information can be modeled as the number of check-ins on the neighbors of a selected POI
- Unvisited POIs near a lot of visited POIs potentially negative
- POI Locality: user's preference on POI is relative to its neighbors

LGM Procedure

LGM Procedure (User's HAL)



LGM Procedure (User's Region)



LGM Procedure (POI Locality)

- POI Locality:

$$p_i^u = 1 - \frac{L_p^u}{|P^u|}$$

- L_p^u denotes the number of neighbors of p_i visited by user u
- Neighbors of p_i: checked-in POIs, whose distance is less than γ meters from the unvisited POIs
- $|P^u|$ is the cardinality of the set of POIs that user u has visite

Logistic Matrix Factorization

- Similar to earlier MF models
- Logistic function to tackle the implicit feedback
- User's preference on a POI

$$P(e_{up}|v_u, l_p, \beta_u, \beta_p) = \frac{exp(v_u l_p^T + \beta_u + \beta_p)}{1 + exp(v_u l_p^T + \beta_u + \beta_p)}$$

Fused Geographical Matrix Factorization

- Fused Local Geographical model to Logistic Matrix Factorization

$$Preference_{up} = P(e_{up}|v_u, l_p, \beta_u, \beta_p) \times \hat{M}(u, p)$$

- Model Flexibility
  - Local Geographical Model can be fused by different models

Experimental Setup

Table: Statistics of the evaluation datasets

Datasets	#Users	#POIs	#Check-ins	Sparsity
Gowalla	5,628	31,803	620,683	99.78%
Foursquare	7,642	28,483	512,523	99.87%

For each user, 20% of most recent check-ins as test set, 10% as tuning set and the remaining 70% as training set. The superscript † denotes significant improvements compared to related baselines ($p < 0.05$).

Performance on Gowalla

Method	Metrics					
	Pre@10	Pre@20	Rec@10	Rec@20	nDCG@10	nDCG@20
LMF	0.0328	0.0272	0.0325	0.0534	0.0167	0.0159
PFMMGM	0.0240	0.0207	0.0258	0.0442	0.0140	0.01440
LRT	0.0249	0.0182	0.0220	0.0321	0.0105	0.0093
PFMPD	0.0217	0.0184	0.0223	0.0373	0.0099	0.0101
LMFT	0.0315	0.0269	0.0303	0.0515	0.0157	0.0150
iGLSR	0.0297	0.0242	0.0283	0.0441	0.0153	0.0145
L-WMF	0.0341	0.0296	0.0351	0.0582	0.0183	0.0178
LGLMF	**0.0373**†	**0.0317**†	**0.0383**†	**0.0629**†	**0.0212**†	**0.0208**†

Performance on Foursquare

Method	Metrics					
	Pre@10	Pre@20	Rec@10	Rec@20	nDCG@10	nDCG@20
LMF	0.0228	0.0189	0.0342	0.0565	0.0136	0.0148
PFMMGM	0.0170	0.0150	0.0283	0.0505	0.0109	0.0126
LRT	0.0199	0.0155	0.0265	0.0425	0.0117	0.0124
PFMPD	0.0214	0.0155	0.0290	0.0426	0.0124	0.0128
LMFT	0.0241	0.0194	0.0359	0.0568	0.0150	0.0161
L-WMF	0.0248	0.0197	0.0387	0.0591	0.0162	0.0174
LGLMF	**0.0266**	**0.0213**†	**0.0424**†	**0.0678**†	**0.0175**	**0.0192**†

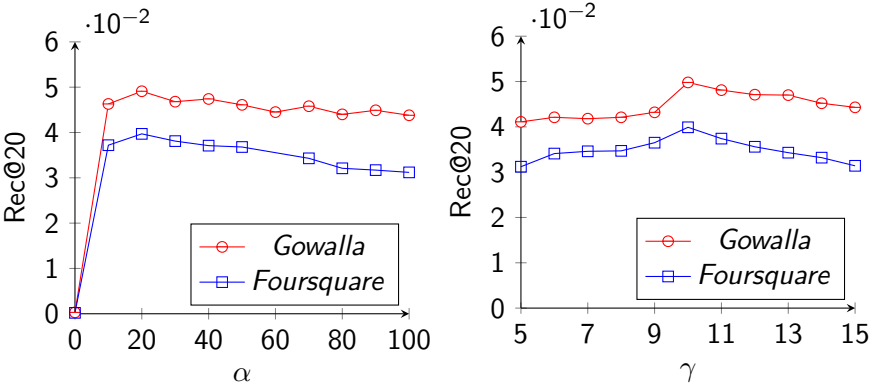
Impact of number of visited POIs (Gowalla)

Method	Rec@20			nDCG@20		
	40%	60%	80%	40%	60%	80%
LMF	0.0205	0.0333	0.0455	0.0058	0.0095	0.0129
PFMMGM	0.0350	0.0376	0.0414	0.0094	0.0107	0.0120
LRT	0.0014	0.0318	0.0300	0.0003	0.0098	0.0083
PFMPD	0.0235	0.0323	0.0346	0.0064	0.0093	0.0088
LMFT	0.0205	0.0321	0.0426	0.0057	0.0088	0.0119
iGLSR	0.0317	0.0357	0.0405	0.0097	0.0105	0.0128
L-WMF	0.0382	0.0435	0.0471	0.0099	0.0110	0.0142
LGLMF	**0.0479**†	**0.0533**†	**0.0580**†	**0.0137**†	**0.0158**†	**0.0187**†

Impact of number of visited POIs (Foursquare)

Method	Rec@20			nDCG@20		
	40%	60%	80%	40%	60%	80%
LMF	0.0180	0.0266	0.0442	0.0048	0.0073	0.0122
PFMMGM	0.0451	0.0466	0.0491	0.0097	0.0107	0.0120
LRT	0.0402	0.0434	0.0438	0.0118	0.0125	0.0125
PFMPD	0.0371	0.0389	0.0412	0.0117	0.0119	0.0124
LMFT	0.0180	0.0288	0.0418	0.0049	0.0077	0.0112
L-WMF	0.0473	0.0492	0.0537	0.0119	0.0131	0.0142
LGLMF	**0.0500**†	**0.0578**†	**0.0660**†	**0.0126**	**0.0154**†	**0.0189**†

Impact of α and γ



(a) Effect of α

(b) Effect of γ

Figure: Effect of different model parameters on the performance of LGLMF

Model Flexibility

Method	Gowalla			Foursquare		
	Pre@20	Rec@20	nDCG@20	Pre@20	Rec@20	nDCG@20
PFMPD	0.0184	0.0373	0.0101	0.0155	0.0426	0.0128
PFMMGM	0.0207	0.0442	0.0144	0.0150	0.0505	0.0126
PFMLGM	**0.0309**†	**0.0588**†	**0.0197**	**0.0198**†	**0.0639**†	**0.0193**
LRT	0.0182	0.0321	0.0093	0.0155	0.0425	0.0124
LRTLGM	**0.0330**†	**0.0616**†	**0.0224**†	**0.0230**†	**0.0717**†	**0.0234**†

Conclusions

- We proposed Local Geographical Logistic Matrix Factorization
- Context-aware POI recommendation
- We considered both users' and locations' point of view
- Effectively modeled the user mobility patterns
- Flexibility to be incorporated by other models

Data and code

Data and code are open source, available at
https://github.com/rahmanidashti/LGLMF

Thanks for your attention!

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